

High Power SOA-integrated DFB Lasers for Co-packaged Optics

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1. Introduction

Exponential growth of AI and machine learning applications drives the demand for interconnect bandwidth in data centers. These applications require the optical interconnect to be low-power and low-latency characteristics at the same time. Co-packaged optics (CPO) and related packaging methods are expected to significantly reduce signal attenuation caused by RF electrical losses. CPO systems typically incorporate silicon photonics (SiPh)-based transmitters along with external laser sources [1]. In this configuration, continuous-wave (CW) high-power operation is required in order to divide the output light into the multiple optical channels. We developed high power 1.3 μm semiconductor optical amplifier (SOA) integrated DFB lasers for external laser sources [2–6]. The basic idea is that the SOA section controls the output power and the DFB section offers stable single-mode operation. In this paper, we will review the recent progress in our development of high-power lasers for CPO applications.

2. Device structure and results

Figure 1 shows a schematic view of the SOA-integrated DFB laser. A light output from the DFB laser is used as a seed light to be amplified by the SOA section. The lasing wavelength and the other properties are mainly determined by the DFB laser. To suppress the facet back reflection, the SOA is terminated with a 7° tilted facet. The straight DFB laser section and the tilted SOA section are connected via a curved waveguide. A high-reflection (HR) and anti-reflection (AR) coatings are formed on the rear side of DFB laser and the front facet of SOA, respectively. DFB laser and SOA sections share identical active layers consisting of InGaAsP quantum wells. To increase the optical volume, the width of the SOA section is expanded beyond transverse single-mode conditions. The fabricated device is evaluated in a *p*-side down configuration. AlN sub-mount is used to bonding the laser chip. Figure 2 shows optical output versus SOA current characteristics measured at various temperatures. A bias current to the DFB laser is fixed at 300 mA. The device exhibits CW optical power exceeding 500 mW at 45°C, and maintains optical power above 300 mW at 70°C. The power conversion efficiency (PCE) at 45°C is 25% with optical power range from 200 to 440 mW. Spectrum measurements show side-mode suppression ratio (SMSR) of greater than 50 dB. Laser linewidth measurements revealed that the device has narrow linewidth of less than 200 kHz.

3. Conclusion

We presented a high-power SOA-integrated DFB laser as an external laser source for CPO application. The results indicate this laser is promising for high-power and power efficient external laser source.

References: [1] J. E. Johnson et al., OFC2022, Tu2D.1 (2022). [2] D. Inoue et al., OFC2023, M4C.4 (2023). [3] D. Inoue et al., OECC2024, TuC2.5 (2024). [4] K. Aoyama et al., SEI Tech. Rev., 83, 40 (2024). [5] D. Inoue et al., ISLC2024, MC2 (2024). [6] D. Inoue et al., OFC2025, W1G.7 (2025).

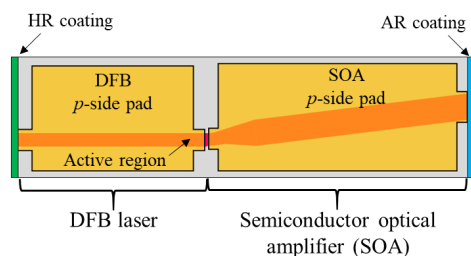


Fig. 1 Top view of an SOA-integrated DFB laser.

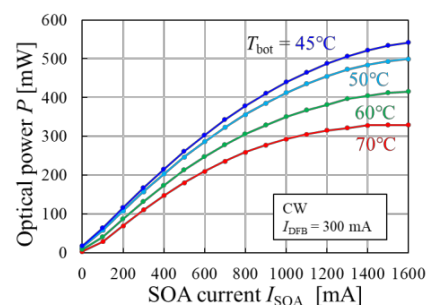


Fig. 2 Light output characteristics.